Portfolio Optimization as a Linear Regression

Overfitting, LASSO and Ridge, and Five Explorations

DBA5106 Foundations of Business Analytics, Project 1

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Group 20

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Code: `DBA5106_Group20_Portfolio_Optimization.ipynb`

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1 Introduction

This project uses a portfolio problem to study overfitting. Every trading day we split our money across 98 assets (portfolios of US stocks sorted by company size and book-to-market). The equally weighted portfolio (EW) puts 1/98 of the money in each asset. The minimum-variance portfolio (MinVar) [2] chooses the weights whose return had the lowest variance over the previous 126 trading days. On the 2nd validation set (2026-01-02 to 2026-07-31), EW reaches a daily Sharpe ratio of 0.111 and MinVar only 0.021. This is in line with DeMiguel et al. [1], who found that none of the 14 models they tested is consistently better than EW. Our question was why the optimized portfolio does worse, and whether LASSO, Ridge and a few reasoned changes can do better.

We worked in six steps, and this report follows the same order:

1. Rebuild the set-up and write MinVar as a linear regression (Sections 2 and 3).
2. Run the baseline (EW, MinVar, LASSO and Ridge) and look for signs of overfitting (Section 4).
3. Use what the baseline showed to plan four explorations, one for each part of the course framework (data, loss, structure, constraints), and fix the rule for the final choice before running them (Section 5).
4. Run the four explorations on LASSO and Ridge, noting what each one taught us (Sections 5.1 to 5.4).
5. Take stock, and add a fifth idea: no negative weights (Section 5.5).
6. Apply the rule to choose one procedure for the August test month (Section 6).

In short, MinVar clearly overfits. LASSO and Ridge remove much of this but still trail EW in 2026, when only the versions with no negative weights beat EW. Our rule picks Ridge with one weight per size group: the best version over 2025 and 2026 together, but weak in 2026 alone.

2 Data and set-up

- **Data.** Daily returns (in percent) from the second table of the 100 Portfolios Formed on Size and Book-to-Market file in Prof. French's Data Library, up to 2026-07-31. We remove ME9 BM10 and BIG HiBM (missing data), which leaves $p = 98$ assets in 10 size groups, from the smallest to the largest companies. No missing values are left in the rows we use, and our code reproduces the three debugging numbers given for the project [3] (row 26,152, the six MinVar weights and the return of 1.393).
- **Rolling window.** For each of the 145 trading days of the 2nd validation set, we fit every portfolio on the 126 trading days before it, keep the weights for that day only and record the return; then the window moves forward by one day. Nothing from the day itself or later is used. The reference Sharpe ratios for EW and MinVar (0.111 and 0.021) match ours only with July included, although the project text ends this set on 2026-06-30.
- **Choosing λ (the 1st validation).** Inside each window, 5-fold cross-validation (CV) over 101 values of λ from 10^{-8} to 10^{8} keeps the λ with the lowest average MSE on the held-out folds, then refits on the whole window. The days are shuffled into the same folds for every version, with a fixed seed; this is fine because daily returns show no clear time pattern ([3], slide 77). We do not standardize, because returns are unitless (slide 78).
- **Score.** The daily Sharpe ratio $\bar{r}/s_r$: the average daily return divided by its standard deviation (SD). EW is our benchmark throughout.
- **A second period.** 145 days is short, so part of any difference may be luck. We therefore ran exactly the same code on 2025 (250 trading days), which also comes before the test month. The test month (August 2026) is not in the data; we treat it as held by the instructor (slide 39).

3 From a portfolio to a regression

We write the weights as $w = w_{\text{EW}} - N\beta$, where N is the 97×97 identity matrix with a row of -1s added below it (so N is 98×97). Every column of N adds up to 0, so the weights add up to 1 whatever β is, and $\beta = 0$ gives EW. In other words, this step removes the sum-to-one constraint, which an ordinary regression cannot handle (slide 76). With R the 126×98 matrix of returns in the window, the target is $y = Rw_{\text{EW}}$ (the daily return of EW) and the features are $X = RN$ (column j is the return of asset j minus the return of the last asset). Then

$$y - X\beta \;=\; R(w_{\text{EW}} - N\beta) \;=\; Rw,$$

so the error of the regression on each day is the portfolio's return on that day. OLS with an intercept makes this error as close to a constant as possible. The intercept takes the average return, so OLS minimizes the variance of the portfolio's return, which gives MinVar. This is why the intercept must be there (slide 75). LASSO and Ridge add a penalty on β; in the scaling of scikit-learn, which our code uses,

$$\text{LASSO: } \min_{\alpha,\beta} \ \tfrac{1}{2n}\|y-\alpha-X\beta\|_2^2+\lambda\|\beta\|_1, \qquad \text{Ridge: } \min_{\alpha,\beta} \ \|y-\alpha-X\beta\|_2^2+\lambda\|\beta\|_2^2.$$

So EW ($\beta = 0$, the most constrained end) and MinVar (OLS, the free end) are the two ends of one path (slide 73): a very large λ gives EW, a very small one gives MinVar, and CV looks for a point in between. With 97 coefficients from 126 days, far below the rule of thumb of 10 observations per feature (slide 16), we expected MinVar to overfit. As β has no intuitive meaning, we always turn it into weights before judging a portfolio.

4 The baseline and what it told us

Table 1 shows the four baseline portfolios. Our EW and MinVar Sharpe ratios match the reference values (0.111 and 0.021). We saw three things.

Table 1: The baseline on the 2nd validation set (2026-01-02 to 2026-07-31, 145 days). The training-window SD and the sum of absolute weights are averages over the 145 days (notebook section 7).

Portfolio	Mean daily return (%)	SD (%)	Sharpe ratio	SD in training window (%)	Validation SD / training SD	Sum of $\|w\|$
EW	0.107	0.958	**0.111**	1.021	0.9	1.0
MinVar	0.018	0.877	0.021	0.183	4.8	12.2
LASSO	0.034	0.555	0.061	0.351	1.6	3.2
Ridge	−0.002	0.581	−0.004	0.309	1.9	4.2

1. MinVar overfits. Inside its 126-day training window, MinVar's daily return had an SD of only 0.18%. On the validation days it was 0.88%, almost five times larger (Figure 1); in 2025 the ratio was 5.6. EW shows no such gap (1.02% and 0.96%). MinVar also uses extreme weights: on an average day its absolute weights add up to 12.2, against 1 for EW, with large positive and negative weights that fit the noise of 126 days (Appendix, Figure A1).

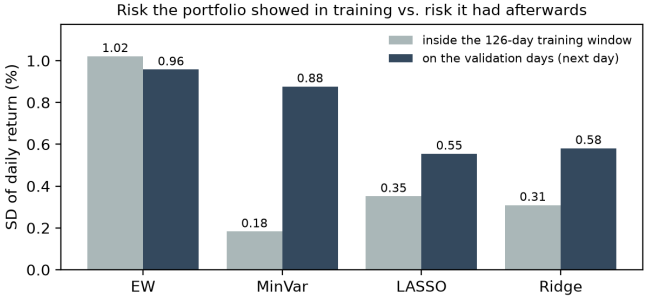


Figure 1: Average SD of each portfolio's daily return inside its 126-day training window, and SD on the 145 validation days of 2026 (notebook section 7).

2. LASSO and Ridge reduce the overfitting, but do not beat EW in 2026. Their absolute weights add up to only 3.2 and 4.2, the validation SD is 1.6 and 1.9 times the training SD instead of 4.8, and they have the lowest risk of the four (SD 0.55% and 0.58%, against 0.96% for EW). On the first trading day of each month in 2026, LASSO set on average 58 of the 97 coefficients to exactly 0, so those assets kept the EW weight of 1/98. But their Sharpe ratios (0.061 and −0.004) stay below EW's, because their average return fell even more than their risk: 0.034% and −0.002% a day, against 0.107% for EW. Table 2 shows part of the reason. Minimizing risk moves the money toward the assets with the lowest SD, the large companies (99% of LASSO's money and 106% of Ridge's), and in 2026 the large companies rose the least: \$1 grew to \$1.11, against \$1.17 for middle and \$1.19 for small companies. Average returns are very hard to predict, and these methods do not try to predict them.

3. The ranking depends on the period. In 2025 EW came last of the four: its Sharpe ratio was 0.043, against 0.088, 0.052 and 0.051 for MinVar, LASSO and Ridge (Table 3). At EW's lowest point

in 2025 (8 April), \$1 invested at the start of the year had fallen to \$0.81 in EW but only to \$0.94 in LASSO and Ridge. In that year, lower risk came with a better result.

Table 2: Where the money went: share of the money in small (size groups 1 to 3), middle (4 to 7) and large (8 to 10) companies, averaged over the first trading day of each month (with negative weights, a share can be below 0% or above 100%), and how the three did, with equal weights inside each (notebook section 10).

	2026			2025		
	Small	Middle	Large	Small	Middle	Large
Share of the money (%)						
EW	30.6	40.8	28.6	30.6	40.8	28.6
MinVar	27.4	−36.4	109.0	−30.9	79.6	51.3
LASSO	8.3	−7.2	98.8	−1.0	10.4	90.6
Ridge	13.8	−20.1	106.3	4.9	4.0	91.1
LASSO, size groups	0.8	−69.0	168.2	17.5	−71.6	154.1
Ridge, size groups	−4.1	−61.2	165.3	16.4	−66.5	150.0
LASSO, no negative weights	0.2	4.8	95.0	5.5	19.0	75.5
Ridge, no negative weights	0.3	5.3	94.5	5.8	17.9	76.3
SD of daily return (%)	1.253	0.985	0.762	1.582	1.427	1.157
Value of \$1 at the end	1.187	1.174	1.108	1.170	1.095	1.132

What this told us. (a) LASSO and Ridge still overfit (validation SD 1.6 to 1.9 times the training SD): we could give each coefficient more data, use fewer coefficients, or restrict β more. (b) CV chooses λ by the MSE, which measures only risk, but we are judged by the Sharpe ratio. (c) One period can mislead, so we judge every version on both, with the rule for the final choice fixed in advance.

5 Five explorations

Before running any exploration, we wrote a short plan with one idea for each part of the framework, the reason for it and what we expected. Each idea changes one thing, is applied to both LASSO and Ridge, and keeps everything else (window, folds, values of λ) the same; only 5.5 uses 21 values of λ instead of 101, which the project instructions allow when fits are slow. We kept the number of versions small on purpose: trying many on the same days and keeping the best would look better than it is. This is the optimizer's curse (slide 39), which the lectures ask us to avoid with a limited number of explorations (Review for Quiz 1, slide 17). We fixed the rule for the final choice at the same time (Section 6). Table 3 shows every version.

Table 3: Every version: daily Sharpe ratio in 2026 (2nd validation set), 2025 (extra check) and both periods (395 days as one series); SD on the 2026 validation days and in the training window (1008 days for 5.1); rank of the 12 candidates over both periods. Bold: final choice (notebook section 8).

		Sharpe ratio			SD in 2026 (%)		
Part	Version	2026	2025	Both	Validation	Training	Rank
Baseline	EW (benchmark)	0.111	0.043	0.062	0.958	1.021	–
	MinVar	0.021	0.088	0.070	0.877	0.183	–
	LASSO	0.061	0.052	0.053	0.555	0.351	6
	Ridge	−0.004	0.051	0.035	0.581	0.309	10
5.1 Data	LASSO, 1008-day window	0.056	0.051	0.052	0.649	0.656	7
	Ridge, 1008-day window	0.051	0.053	0.052	0.626	0.653	8
5.2 Loss	LASSO, λ by Sharpe	0.018	0.004	0.007	0.671	0.301	12
	Ridge, λ by Sharpe	−0.012	0.053	0.034	0.710	0.269	11
5.3 Structure	LASSO, size groups	0.029	0.120	0.093	0.675	0.607	2
	Ridge, size groups	**0.025**	**0.132**	**0.100**	**0.679**	**0.609**	**1**
5.4 Constraints	LASSO, one-SE λ	0.095	0.052	0.061	0.545	0.418	5
	Ridge, one-SE λ	0.034	0.056	0.049	0.562	0.389	9
5.5 Constraints	LASSO, no negative weights	0.130	0.063	0.078	0.558	0.538	4
	Ridge, no negative weights	0.134	0.067	0.081	0.555	0.539	3
	MinVar, no negative weights	0.130	0.065	0.079	0.558	0.538	–

5.1 Data: a longer window

Why. 97 coefficients from 126 days is far below the rule of thumb, which asks for about 970 days here. *What we changed.* The window length. We had planned 252, 504 and 1008 days and ran all three; we report 1008 days (about four years), the length the rule of thumb points to, and the other two give the same final choice. *What we expected.* Less noisy coefficients and less overfitting; the risk is that older days behave differently from 2026. *What happened.* The overfitting gap closed: for LASSO the SD was 0.66% in training and 0.65% on the validation days (Ridge: 0.65% and 0.63%). The Sharpe ratio improved for Ridge in 2026 (-0.004 to 0.051) but not for LASSO (0.061 to 0.056), and in 2025 it hardly moved (0.052 to 0.051 for LASSO, 0.051 to 0.053 for Ridge). *What we learned.* More data closed the gap between training and new days, as the rule of thumb predicts. But a smaller gap is not the same as a better portfolio: the risk on new days was higher than with 126 days (0.63% to 0.65% against 0.55% to 0.58%), possibly because older days behave differently, the risk we expected.

5.2 Loss: choosing λ by the Sharpe ratio

Why. The project's validation loss is minus the Sharpe ratio, but CV chose λ by the MSE, which only measures how much the portfolio's return moves around its average. *What we changed.* For each λ and fold, we fit on the other four folds and compute the Sharpe ratio of the portfolio's returns on the held-out fold; we keep the λ with the highest average. *What we expected.* We were not sure. It could keep more of EW's return, but each fold has only about 25 days, so the chosen λ might follow noise. *What happened.* It was worse: in 2026 LASSO fell from 0.061 to 0.018 and Ridge from -0.004 to -0.012, and over both periods they were the two weakest candidates. It chose much smaller values of λ (median 0.003 instead of 0.025 for LASSO), so the weights moved back toward MinVar (absolute weights adding up to about 7 to 8) and the overfitting gap grew. *What we learned.* A Sharpe ratio measured on 25 days has a standard error of about $1/\sqrt{25} = 0.2$, far larger than the differences we care about, so CV followed noise in average returns. The MSE mostly measures risk, which is much easier to estimate.

5.3 Structure: one weight per size group

Why. Assets of a similar company size tend to move together, so many of the 97 separate coefficients may only fit noise. With one return per size group we have 9 coefficients, 14 days per coefficient, which meets the rule of thumb without going back in time (the problem in 5.1). *What we changed.* We average the assets of each of the 10 size groups, fit the same LASSO and Ridge on these 10 returns, and share each group's weight equally among its assets. *What we expected.* Much less overfitting; the risk is that equal weights inside a group remove useful differences (more bias). *What happened.* The overfitting almost disappeared (Ridge: 0.61% in training, 0.68% on the validation days). But the two years disagree: in 2025 these were the best versions of all (0.120 for LASSO and 0.132 for Ridge, against 0.043 for EW); in 2026 they were in the lower half (0.029 and 0.025). Table 2 shows why: with only 10 weights, the regression was positioned the same way in both years, with 150% to 168% of the money in large companies and -61% to -72% in middle-sized ones. In 2025 the middle-sized companies rose the least (\$1 became \$1.09, against \$1.13 for large ones), so this helped. In 2026 they rose more than large ones (\$1.17 against \$1.11), so the same position hurt. *What we learned.* Fewer free coefficients means much less overfitting, as expected. But the result then depends on one difference between size groups, and that difference changed from year to year.

5.4 Constraints: a larger λ (the one-standard-error rule)

Why. The λ with the lowest CV error, $\lambda^\star$, is itself estimated from five small folds, so it is noisy. A larger λ means a smaller region for β and a portfolio closer to EW. *What we changed.* We take the largest λ whose average CV error is at most the lowest average CV error plus one standard error (the SD of the five fold errors at $\lambda^\star$, divided by $\sqrt{5}$). *What we expected.* Slightly more risk, but more of EW's return. *What happened.* In 2026 it helped both methods: LASSO rose from 0.061 to 0.095 and Ridge from -0.004 to 0.034. In 2025 it made little difference (LASSO 0.052 in both cases, Ridge 0.051 to 0.056). The chosen λ was about two to three times larger (median 0.052 instead of 0.025 for LASSO, 83 instead of 28 for Ridge), and the gap between validation and training SD fell (1.3 instead of 1.6 for LASSO). *What we learned.* When the data are this noisy, leaning toward a simpler model helped in

2026 and made little difference in 2025, as the bias-variance tradeoff would suggest.

Taking stock after four explorations. Every idea that limited the freedom of β (more days per coefficient, fewer coefficients, a larger λ) made the overfitting gap smaller; the new loss made it larger. Only the larger λ helped both LASSO and Ridge in 2026 without hurting 2025. Every version still used negative weights, MinVar most of all (absolute weights adding up to 12.2). We therefore added one idea from our plan that we had first left out to keep the number of versions small: no negative weights.

5.5 Constraints: no negative weights

Why. In the lecture's age-guessing example, constraints that were not exactly true still lowered the error, because they reduce freedom and so the variance (Loss Function and Linear Regression, slide 14). The true best portfolio may need some negative weights, but without them an estimated portfolio cannot take large positions that only fit noise. *What we changed.* We solve the same LASSO and Ridge problems as in Section 3 with one extra rule, $w = w_{\text{EW}} - N\beta \geq 0$. Because scikit-learn cannot add this rule, we use the `cvxpy` package with the same scaling, and λ is chosen by the same CV on the same folds, with the rule inside every fold. It is still the regression: we solve for β and use $w = w_{\text{EW}} - N\beta$; we never solve the minimum-variance problem directly (slide 80). A version with $\lambda = 0$ (Table 3: MinVar, no negative weights), solved the same way, is only a reference, to separate the effect of the rule from that of the penalty. *What we expected.* Like the penalty, the rule limits how far β can move, so it should also reduce overfitting. *What happened.* These were the best versions in 2026 (0.134 for Ridge and 0.130 for LASSO, against 0.111 for EW) and above EW in 2025 too (0.067 and 0.063, against 0.043). They are the only versions above EW in both periods. Over both periods together they came third and fourth (0.081 and 0.078). The overfitting gap disappeared (Ridge: 0.54% in training, 0.56% on the validation days). The $\lambda = 0$ version scored almost the same (0.130 in 2026, 0.065 in 2025), which suggests that most of the effect comes from the rule itself, not from the penalty. The cost is that the money is spread over very few assets: on 2026-01-02 each version held only 5 of the 98 assets with a weight above 0.001 (notebook section 5.5; Figure A1 shows the Ridge version), about 0.4 of the money was in a single asset on an average day, and about 95% of the money was in large companies in 2026 (Table 2). *What we learned.* A simple constraint that is not true in general did more against overfitting than the penalty alone, but it leaves the money in very few assets. A similar version (other folds and values of λ) had been tried before this exploration was planned.

6 The final choice

The rule, fixed before the explorations were run. Among the LASSO and Ridge versions (12 in Table 3), choose the one with the highest Sharpe ratio over 2025 and 2026 together, computed on all 395 daily returns as one series (not the average of the two Sharpe ratios). The list of versions changed after that (5.5 was added, and 5.1 reports one of its three windows; with all three the choice is the same), but the rule did not. MinVar versions are not candidates, since the final procedure is built with LASSO or Ridge; EW is the benchmark.

The choice: Ridge with one weight per size group (Section 5.3): a 126-day window, Ridge on the returns of the 10 size groups, λ by 5-fold CV over the 101 values, and each group's weight shared equally among its assets. Its Sharpe ratio over both periods is 0.100, against 0.062 for EW. The next candidates are LASSO with size groups (0.093) and Ridge with no negative weights (0.081). For August we would run this procedure without any change.

Uncertainty. We are not very confident in this choice, for three reasons. First, the lead comes from 2025 (0.132). In 2026 this version reached only 0.025, below EW, and on 2026 alone the same rule would have picked Ridge with no negative weights (0.134); Figure 2 shows both years. Second, a daily Sharpe ratio over n days has a rough standard error of $1/\sqrt{n}$, about 0.05 for our 395 days: more than the lead over EW (0.100 against 0.062). Third, both periods were used to choose, so neither is an untouched test, and 0.100 is probably a little optimistic. We still keep the rule: changing it after seeing the results would be the optimizer's curse again.

For description only, three further measures (Appendix, Table A1) give a similar picture over both periods: every LASSO and Ridge version had a smaller maximum drawdown than EW (the largest fall of \$1 below its earlier high: -11.7% to -14.9%, against -22.3%), and the chosen version reached the

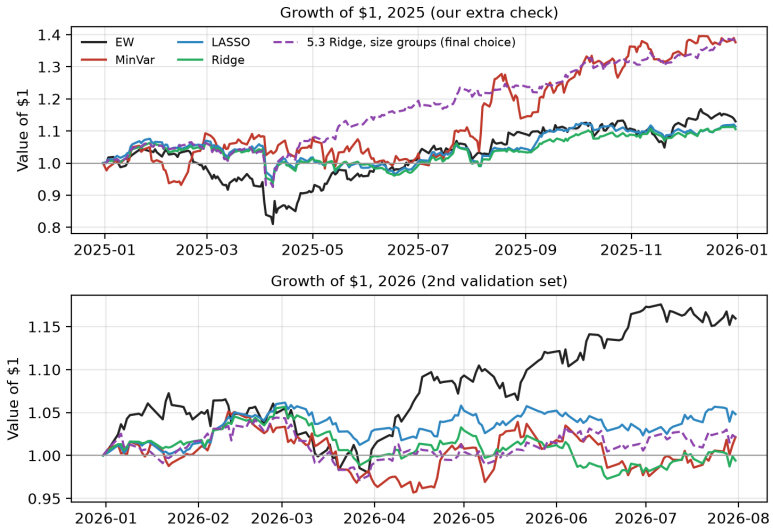


Figure 2: Growth of $1 in 2025 (our extra check, top) and in 2026 (the 2nd validation set, bottom) for the baseline portfolios and the final choice (dashed line; notebook section 8).

highest value of $1 among the 12 candidates (1.43, against 1.33 for EW).

7 Limitations and what we learned

Limitations. The periods are short and most differences between versions are within noise; the chosen version was behind EW in 2026 and may well be behind it in August. We ignored trading costs, although the weights are rebuilt every day (MinVar, with its large weights, would be affected most). All results use one version of French's data (the library revises past data over time; our copy ends on 2026-07-31).

What we learned.

1. Overfitting can be seen directly, by comparing the risk a portfolio showed in training with its risk on the next days. Every way of limiting the freedom of β narrowed this gap (more days, fewer coefficients, a larger λ, no negative weights): LASSO and Ridge are one kind of constraint among several.
2. Lower risk does not guarantee a higher Sharpe ratio. Average returns are hard to predict, and how small, middle and large companies did, which changed between 2025 and 2026, decided much of the ranking.
3. A loss has to be estimable as well as relevant: the Sharpe ratio is what we care about, but 25 days are too few to estimate it. How we choose also matters: we kept to a small number of reasoned ideas and to a rule fixed in advance.

AI declaration

As the course encourages, we used Claude Opus 5.5 (Anthropic) to help write, debug and re-check the code and to draft and edit this report. We chose the explorations and the final-choice rule, checked every number against our notebook, and take full responsibility for the work.

References

[1] DeMiguel, V., Garlappi, L. and Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5), 1915–1953.

[2] Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91.

[3] Zhao, L. (2026). *Portfolio Optimization* (Project 1 instructions) and DBA5106 lecture slides (*Loss Function and Linear Regression*; *Overview of Project 1*; *Review for Quiz 1*), National University of Singapore. Slide numbers in the text refer to *Overview of Project 1* unless another deck is named.

Appendix

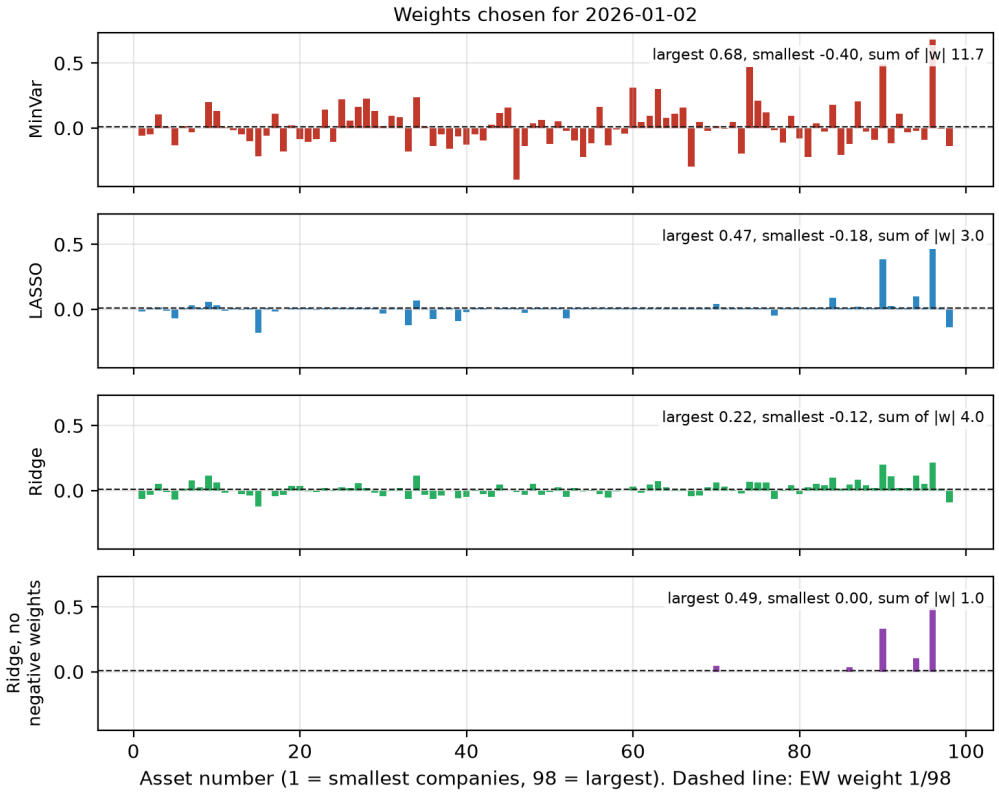


Figure A1: Weights chosen for 2026-01-02 by MinVar, LASSO, Ridge and Ridge with no negative weights, all from the same 126 days. Dashed line: the EW weight 1/98 (notebook section 7).

Table A1: Three extra measures, for description only (the final choice uses only the Sharpe ratio): the largest fall of the value of $1 below its highest earlier value, the highest value of $1, and the largest absolute weight averaged over the days (notebook section 9). Bold: our final choice.

	Both periods (395 days)			2026 (145 days)		
Version	Max drawdown (%)	Highest value of $1	Largest weight	Max drawdown (%)	Highest value of $1	Largest weight
EW	−22.3	1.33	0.01	−8.6	1.18	0.01
MinVar	−11.6	1.45	0.79	−9.0	1.05	0.57
LASSO	−12.3	1.18	0.42	−4.6	1.06	0.42
Ridge	−11.9	1.17	0.22	−7.9	1.06	0.21
LASSO, 1008-day window	−12.7	1.22	0.32	−4.9	1.10	0.29
Ridge, 1008-day window	−12.0	1.21	0.26	−4.6	1.09	0.23
LASSO, λ by Sharpe	−11.7	1.08	0.52	−5.6	1.05	0.45
Ridge, λ by Sharpe	−14.1	1.17	0.39	−7.2	1.04	0.38
LASSO, size groups	−13.4	1.39	0.15	−6.9	1.04	0.11
Ridge, size groups	**−13.1**	**1.43**	**0.14**	**−6.9**	**1.04**	**0.11**
LASSO, one-SE λ	−12.8	1.22	0.40	−4.6	1.09	0.40
Ridge, one-SE λ	−14.1	1.19	0.14	−6.1	1.05	0.15
LASSO, no negative weights	−14.9	1.29	0.42	−4.1	1.12	0.39
Ridge, no negative weights	−14.9	1.30	0.38	−4.1	1.12	0.37
MinVar, no negative weights	−14.8	1.29	0.43	−4.1	1.12	0.39